

School of Arts and Sciences

Department of Computer Science and Information Technology

Summer 2024 – 2025

CSCI390 Web Programming

Class Day	:	_____	Class Time:	_____	Room:	_____
Instructor	:	_____				
E-mail	:	_____				
Office	:	_____	Office Hours:	_____	Extension No:	_____

I. Course Description

This course presents the fundamentals of web programming and design at client side and server side. At first, the course introduces students to Hyper Text Markup Language (HTML) which is the basic language used to create properly structured web pages. Students then learn Cascading Style Sheet, which allows them to design the content of web pages. With the variety of devices from which websites are accessed today, designing a responsive website became a must. Therefore, students learn how to structure the web pages content to be displayed responsively on different screen dimensions. To create dynamic websites, students learn the most popular scripting language JavaScript.

The course will introduce the principles necessary to develop server-side scripts used to build dynamic websites and applications using PHP script language to connect to the server database and display the data from the server.

II. Course Objectives

CO-1: Learn to structure web pages using HTML, CSS

CO-2: Learn the most popular scripting language JavaScript

CO-3: Introducing the server-side scripting language PHP

CO-4: Test the web by connecting and displaying the database in the web page using PHP and MySQL

III. Course Learning Outcomes

CLO-1: Create a properly structure web site

CLO-2: Design a responsive web site

CLO-3: Distinguish PHP as a server-side scripting language

CLO-4: Create a dynamic web site using front-end JavaScript

CLO-5: Introducing a dynamic data-driven web application using PHP and MySQL

IV. Mapping to program outcomes

Web Programming	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1			X	X					
CLO-2				X	X				
CLO-3			X						
CLO-4				X	X	X			
CLO-5				X		X			

V. Mapping course outcomes to Test

Web Programming	Mapping to Test	
	Q1 (50 pts)	Q2 (50 pts)
CLO-1	50 pts	50 pts
CLO-2		
CLO-3		
CLO-4		
CLO-5		

VI. Mapping course outcomes to Midterm

Web Programming	Mapping to Midterm	
	Q1 (50 pts)	Q2 (50 pts)
CLO-1	50 pts	
CLO-2		
CLO-3		
CLO-4		50 pts
CLO-5		

VII. Mapping course outcomes to project

Web Programming	Mapping to Project			
	Design	Responsive Front -end Scripting	Server script manipulating Db	Implementing-hosting
CLO-1	30 pts			
CLO-2		20 pts		
CLO-3				
CLO-4		20 pts		
CLO-5			20pts	10 pts

VIII. Mapping course outcomes to Final exam

Web Programming	Mapping to Final exam		
	Q1 (20 pts)	Q2 (50 pts)	Q3 (30 pts)
CLO-1	20 pts		
CLO-2			
CLO-3			
CLO-4		50 pts	
CLO-5			30 pts

IX. Topics Covered

- Create a proper web site
 - web site Use an HTML to create a web page.
 - Apply CSS code for formatting a properly structured web page.
 - Manipulate the document object model to create dynamic web pages using JavaScript.
 - Design responsive web pages.
- Test (Use) JavaScript events for developing the web pages.
- Introducing the server-side script language PHP.
- Accessing the database on the server side and display the data on the web pages using PHP and MySQL.

X. Prerequisite:
CSCI300, CSCI335

XI. Course Material: Book or handouts
Handouts are to be distributed in class.
Reference: www.W3Schools.com

XII. *Software required (for computer-lab courses)*
XAMP, Visual studio code

XIII. Course Assessment

Test	20%
Midterm	30%
Project	20%
Final Exam	30%

Withdrawal Deadline: Week 4

XIV. Detailed Schedule

Week	Book Reference	Topic Details
Week 1 30 June-3 July	Lecture 1	Handout Syllabus: reading and explaining. ○ Explaining the course syllabus. ○ History of Internet and World Wide Web. ○ World Wide Web Consortium (W3C). ○ Difference between markup languages, scripting languages and programming languages. ○ What is and why to use HTML, CSS and JavaScript. ○ Difference between interpreter and compiler. ○ Difference between static and dynamic websites. ○ Difference between client-side and server-side programming. ○ Why we need responsive websites. ○ HTML versions and document type declaration (doctype). ○ Domain name, hosting, DNS. ○ Explore Visual Studio Code IDE ▪ How to start a new web project. ▪ How to open an existing web project. ▪ How to add extensions (live server,...). ○ Main elements: DOCTYPE, html, head and body sections.
	Lecture 2	Handout Getting familiar with the IDE and learning basic HTML elements. ○ Main elements: DOCTYPE, html, head and body sections. ○ Heading elements, paragraph, division, span, comment. ○ Formatting elements: italic, underline, bold, small, superscript, subscript, markup, strikethrough, quote, blockquote, pre-formatted text, line break, horizontal break, tooltip (title attribute), Special characters: non-breaking space, copyright... &nbsp; &copy;
	Lecture 3	Handout HTML Lists, image element and Links ○ Unordered list, ordered list, description list, nested lists, changing list style type in html. ○ Image with specification of alternative text, height and width in html. ○ Hyperlinks (anchors) on text and images to open local html pages, images, default email application (mailto) and urls, on same tab and on new tabs (target attribute).
	Lecture 4	Handout HTML tables ○ Basic table with table rows and table data. ○ Complex table with head, body and foot sections, caption, column span and row span. ○ HTML DIV - definition and usage HTML form elements. ○ Form: label, input types (text, email, password, number, date, radio, checkbox) and attributes (name, value, read only, disabled, for, size, max length), button (submit, reset), select element with options, text area.

Week 2 7-10 July	Lecture 1	Handout	HTML form elements ○ Briefly explaining about form validation and form method and action without applying them in code. HTML5 media and structural elements. ○ HTML5 media: picture, video and audio elements with auto play and controls attributes. ○ HTML5 new structural elements: main, header, footer, nav, article, section, aside, ...
	Lecture 2	Handout	Introducing Cascading Style Sheet (CSS) and text styling. ○ CSS introduction and syntax. ○ Demonstration of the different styling approaches: inline, embedded and external styling. ○ Explaining when each styling approach is used, and why external styling is the best practice approach. ○ Text styling: font families (serif, sans-serif, monospace), font size, font style (italic), font weight (bold), font shorthand, text transform (uppercase, lowercase, capitalize), text decoration (none, line, style, color), text align (left, center, right, justify), vertical align (top, bottom, middle), line height, text indent, word spacing, letter spacing.
	Lecture 3	Handout	CSS colors, backgrounds, borders, outlines and list styles. ○ Color (named, rgb, rgba). ○ Background: color, image, repeat (horizontal, vertical, no repeat), attachment. ○ Border: style, color, width, side (top, right, bottom, left), radius, border shorthand. ○ Outline: style, color, width, side (top, right, bottom, left), radius, border shorthand. ○ List style: type (none, square, circle, disc, decimal, upper-alpha, lower-alpha, upper-roman, lower-roman), image (url).
	Lecture 4	Handout	CSS Box model, element display and positioning. ○ Explaining box model concept. ○ Height, width and max width of elements. ○ Element display: none, inline, inline block, block. ○ Margin and padding: side (top, right, bottom, left), auto, shorthand. ○ Box sizing. ○ Position: static, relative, fixed, absolute, sticky.

Week 3 14-17 July	Lecture 1	Handout	CSS selectors, image and filters. ○ Element selectors, id selector, class selector, attribute selectors, combining selectors... ○ Image with specification of alternative text, height and width in html. ○ Filters: anchor filters (hover, visited, active), element filters (first-child, last-child, nth-child, not, first-letter, first-line). CSS Table styling and CSS float. ○ Styling a complex table: border, border collapse, height, width, padding, text alignment, row hover filter. ○ Floating content on several columns (float, overflow, clear).
	Lecture 2	Handout	Introduction to responsive web design with CSS Grid. ○ CSS grid container and items. – Display grid, grid template columns and rows (auto and fixed size values), grid gap on columns and rows, grid area, align content and justify content (start, center, end, space around, space evenly, space between). ○ CSS3 elements and measurement units. ○ Media queries (min-width, max-width).
	Lecture 3	Handout	Responsive Web Design ○ Bootstrap introduction, concepts & setup ○ Bootstrap Navbar ○ Bootstrap carousel ○ Solving Exercises
	Lecture 4	Handout	Test – HTML and CSS
Week 4 21-24 July	Lecture 1	Handout	Introduction to JavaScript using exercises ○ JavaScript basic syntax ○ Demonstration of the different scripting approaches: inline, embedded and external scripting. ○ Demonstrating window prompt and alert, document write and console logging. ○ JavaScript variables and data types: explain type of (undefined, number, string, boolean, object) and how we can assign values of different types to same variable.

	Lecture 2	Handout	Introduction to JavaScript using exercises ○ JS Statements, operators, conditional statements, Loops, functions, arrays. ○ Brief introduction to Math and Date classes methods. Manipulating Document Object Model. ○ Accessing elements by id, class and tag names. ○ Accessing elements using query selectors.
	Lecture 3	Handout	Manipulating Document Object Model. ○ Changing css style. ○ Navigating between nodes: parent node, child nodes, first child, last child, nth child ○ Add, Remove elements. ○ JavaScript events (mouse events, key events, click, double click, load)
	Lecture 4	Handout	Midterm exam
Week 5 28-31 July	Lecture 1	Handout	PHP coding. ○ PHP syntax, how to embed within html document, General structure, Identifiers, Variables (types, values). ○ PHP concatenation, Constants, Scope, Operators, Expressions ○ Global, local variables. ○ Running a PHP file on localhost using PHP, XAMP ○ Conditional statements If, else Switch ○ Iteration: For loop, While , Do while ○ Array declaration ○ Include / require ○ Briefly explain: \$_GET, \$_POST ○ Access Form elements: Form Submission ○ Handling forms using POST , GET.
	Lecture 2	Handout	Setting up MySQL database using phpMyAdmin ○ Setting up Database, Creating a database ○ Import, export database ○ Opening and closing Connections. ○ Creating connection (connect MySQL DB to php pages) ○ Handle the connection, retrieving data from the server ○ Display data in the web page.
	Lecture 3	Handout	Project submission.
	Lecture 4	Handout	Final Exam.

XV. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4	D+	72	2.2	B+
65	1.5		73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	